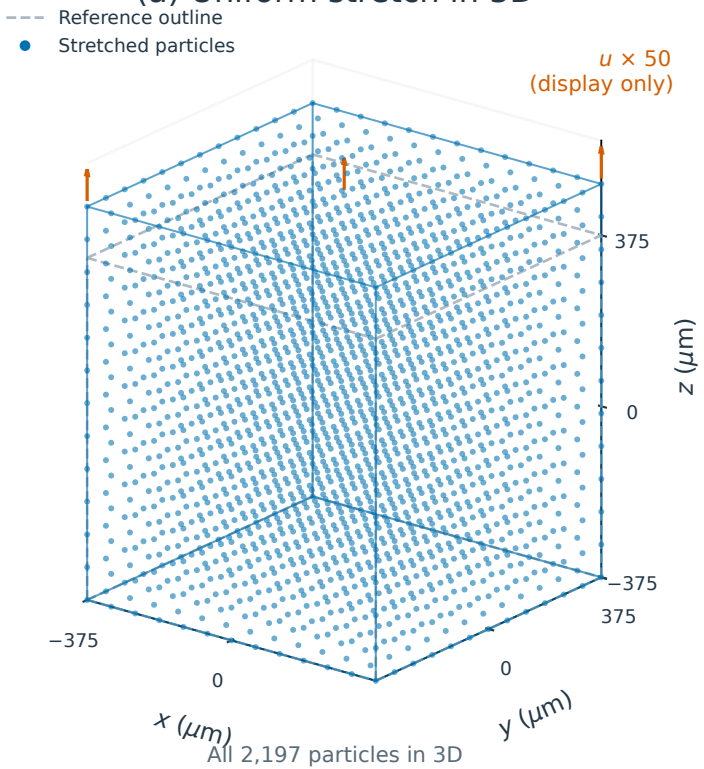
2. Test the Force Calculation

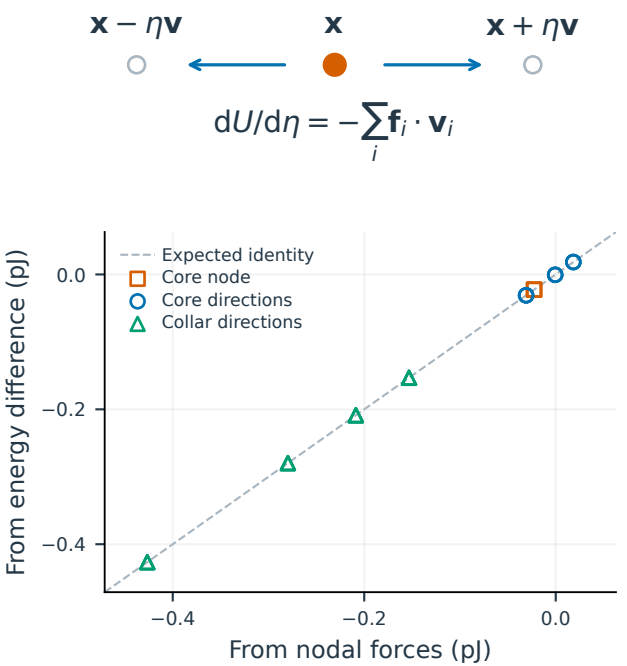
Archived L075 affine / force checks and spectral material-point tests

(a) Uniform stretch in 3D



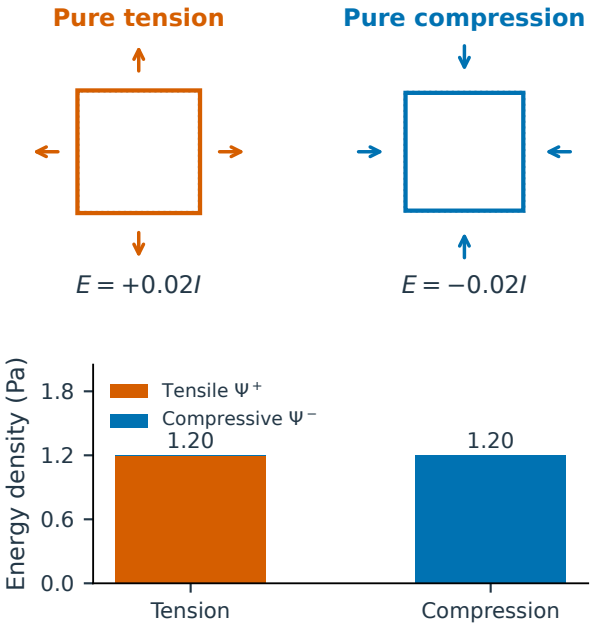
All 2,197 particles in 3D

(b) Energy–force check



All 24 saved comparisons; coincident points overlap.

(c) Stretching versus squeezing



Phase driver: $Y = \Psi^+$

Tension: 1.20 Pa Compression: 0 Pa

Affine deformation reproduced

Actual strain: 0.3%; extension: 2.25 μm
Max gradient error: 4.66e-15

Energy and forces agree

8 directions × 3 steps; max error 9.49e-06

Only tensile energy drives phase

Compression still stores elastic energy

Affine accuracy does not establish force equilibrium. Perturbation checks cover the saved directions, not every particle independently.

Tensile phase driving is a constitutive rule; these verification results do not establish physical clot tearing.